

Guide to Typst

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1 | Introduction

The goal of this document is to have the most common used elements for the markup language **typst** readily available. A detailed documentation can be found on their website: <https://typst.app/docs> It is to note that these are **my** most common used elements. For some elements custom templates are needed:

- [codelst](#)
- [glossarium](#)
- [tablex](#) (*for legacy tables only*)
- all files and folders in the **/00-templates/** folder such as
 - **icons/*.svg**
 - **scripts/*.bash**
 - **syntax/*.sublime-syntax**
 - **boxes.typ**
 - **constants.typ**
 - **helpers.typ**
 - **i18n.json**
 - **items.typ**
 - **karnaugh.typ**
 - **page-*.typ**
 - **sections.typ**
 - **template-***

2 | Installation

2.1 With cargo

If you use already the **rust** programming language then you can use rust to install the latest toolchain.

```
# install rust and cargo
curl https://sh.rustup.rs -sSf | sh

# install typst
cargo install --git https://github.com/typst/typst
```

2.2 MacOS

On MacOS you can use **homebrew**

```
# install homebrew
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"

# install typst
brew install typst
```

2.3 Linux

In Linux you can use the commonly available package manager

```
brew install typst
pacman -S typst
xbps-install typst
sudo apt-get install typst
```

2.4 Windows

On Windows you can use **chocolatey**. See: <https://chocolatey.org/install>

```
# install chocolatey
# ensure to use a administrative powershell
Set-ExecutionPolicy Bypass -Scope Process -Force;
[System.Net.ServicePointManager]::SecurityProtocol =
[System.Net.ServicePointManager]::SecurityProtocol -bor 3072; iex ((New-Object
System.Net.WebClient).DownloadString('https://community.chocolatey.org/install.ps
1'))

# install typst
choco install typst
```

3 | Formatting

3.1 Markup

Name	Example	Raw
Singleline Comment		//
Multiline Comment		/* */
Paragraph break		blankline
Line break		\
bold	bold	*bold*
italic	<i>italic</i>	_italic_
monospaced	monospaced	`monospaced`
math	$x = 1$	\$x=1\$
lowercase	lower	#lower("LoWeR")
uppercase	UPPER	#upper("UpPeR")
smallcaps	SMALLCAPS	#smallcaps("SmallCaps")
smartquote	„test“	#smartquote()test#smartquote()
overline	<u>overline</u>	#overline("overline")
underline	<u>underline</u>	#underline("underline")
strike	strike	#strike("strike")
sub	Text _{sub}	Text#sub("sub")
super	Text ^{super}	Text#super("super")
Label		<label>
Reference		@label

3.2 Page Formatting

```
#pagebreak() // pagebreak
#parbreak() // parbreak
\ // linebreak
```

3.3 Space

A

B

A #h(5cm) B,

C

C #v(0.2cm) D

D

3.4 Text Formatting

For the custom textsizes and colors you need to import:

```
#import "/01-tail/constants.typ": *
```

Name	Example	Raw
Sizes	8pt text tiny text	<code>text(8pt, "8pt text")</code> <code>text(tiny "tiny text")</code>
	9pt text smaller text	<code>text(9pt, "9pt text")</code> <code>text(smaller "smaller text")</code>
	10pt text small text	<code>text(10pt, "10pt text")</code> <code>text(small "small text")</code>
	11pt text normal text	<code>text(11pt, "11pt text")</code> <code>text(normal "normal text")</code>
	14pt text large text	<code>text(14pt, "14pt text")</code> <code>text(large "large text")</code>
	16pt text larger text	<code>text(16pt, "16pt text")</code> <code>text(larger "larger text")</code>
	24pt text huge text	<code>text(24pt, "24pt text")</code> <code>text(huge "huge text")</code>
	36pt text huger text	<code>text(36pt, "36pt text")</code> <code>text(huger "huger text")</code>
	Fira Sans	<code>text(font:"Fira Sans", "Fira Sans")</code>
	Fira Mono	<code>text(font:"Fira Mono", "Fira Mono")</code>
Types	Source Sans Pro	<code>text(font:"Source Sans Pro", "Source Sans Pro")</code>
	Arial	<code>text(font:"Arial", "Arial")</code>
	Times New Roman	<code>text(font:"Times New Roman", "Times New Roman")</code>
Alignment	start	<code>align(start){start}</code>
	end	<code>align(end){end}</code>
	left	<code>align(left){left}</code>
	center	<code>align(center){center}</code>

Name	Example	Raw
	right	<code>align(right){right}</code>
	top	<code>align(top){top}</code>
	horizon	<code>align(horizon){horizon}</code>
	bottom	<code>align(bottom){bottom}</code>
	center + horizon	<code>align(center + horizon){center + horizon}</code>
Colors	black	<code>#text(fill:black)[black]</code>
	red	<code>#text(fill:red)[red]</code>
	green	<code>#text(fill:green)[green]</code>
	blue	<code>#text(fill:blue)[blue]</code>
	purple	<code>#text(fill:purple)[purple]</code>
	gray-80	<code>#text(fill:gray-80)[gray-80]</code>
	gray-70	<code>#text(fill:gray-70)[gray-70]</code>
	gray-60	<code>#text(fill:gray-60)[gray-60]</code>
	gray-50	<code>#text(fill:gray-50)[gray-50]</code>
	gray-40	<code>#text(fill:gray-40)[gray-40]</code>
	gray-30	<code>#text(fill:gray-30)[gray-30]</code>
	gray-20	<code>#text(fill:gray-20)[gray-20]</code>
	gray-10	<code>#text(fill:gray-10)[gray-10]</code>
	hei-orange	<code>#text(fill:hei-orange)[hei-orange]</code>
	hei-blue	<code>#text(fill:hei-blue)[hei-blue]</code>
	hei-pink	<code>#text(fill:hei-pink)[hei-pink]</code>
	hei-yellow	<code>#text(fill:hei-yellow)[hei-yellow]</code>
	hei-green	<code>#text(fill:hei-green)[hei-green]</code>
	spl-green	<code>#text(fill:spl-green)[spl-green]</code>
	spl-blue	<code>#text(fill:spl-blue)[spl-blue]</code>
	spl-pink	<code>#text(fill:spl-pink)[spl-green]</code>
	color-info	<code>#text(fill:color-info)[color-info]</code>
	color-idea	<code>#text(fill:color-idea)[color-idea]</code>
	color-warning	<code>#text(fill:color-warning)[color-warning]</code>
	color-important	<code>#text(fill:color-important)[color-important]</code>
	color-fire	<code>#text(fill:color-fire)[color-fire]</code>
	color-rocket	<code>#text(fill:color-rocket)[color-rocket]</code>
	color-todo	<code>#text(fill:color-todo)[color-todo]</code>

Name	Example	Raw
	<code>code-bg</code>	<code>#text(fill:code-bg)[code-bg]</code>
	<code>code-border</code>	<code>#text(fill:code-border)[code-border]</code>

4 | Elements

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4.13	Word/Character Count	28

4.1 Headings

```
= Heading 1
== Heading 1.1
=== Heading 1.1.1
==== Heading 1.1.1.1
...
```

4.2 Lists

- First
- Second
- Third

- First
- Second
- Third

- First
 - Second
 - Third

- First
 - Second
 - Third

- First
- Second
- Third

```
- First
- Second
- Third
```

- First
- Second
- Third

```
list(
  [First],
  [Second],
  [Third],
)
```

1. First
 1. Second
2. Third

```
+ First
+ Second
+ Third
Text
4. Fourth
+ Fifth
```

Text

4. Fourth
5. Fifth

```
+ First
#set enum(numbering: "a")
+ Second
+ Third
Text
4. Fourth
+ Fifth
```

1. First
 - a) Second
2. Third

Text

4. Fourth
5. Fifth

4.3 Custom Lists

```
#import "/00-templates/items.typ": *
```

- ☒ item-list
- ☒ item-checkbadge
- ☐ item-circle
- ☐ item-square
- ☒ item-checkcircle
- ☒ item-checksquare
- ✓ item-check
- ☐ item-file
- ☐ item-folder
- ☒ item-xcircle
- ☒ item-xsquare
- × item-x

```
#item-list()[item-list]
#item-checkbadge()[item-checkbadge]
#item-circle()[item-circle]
#item-square()[item-square]
#item-checkcircle()[item-checkcircle]
#item-checksquare()[item-checksquare]
#item-check()[item-check]
#item-file()[item-file]
#item-folder()[item-folder]
#item-xcircle()[item-xcircle]
#item-xsquare()[item-xsquare]
#item-x()[item-x]
```

4.4 Minitoc

The `minitoc` is specific for this template and allows to show a TOC between two labels. The `minitoc` is also used within the `#add-chapter` function if `after` and `before` labels are defined.

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4.8.2	Option Style	21
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4.11	Exam Reminder	25
4.12	Exercise Message	27
4.13	Word/Character Count	28

```
#minitoc(after: <sec:elem>, before: <sec:ref>)
```

```
#add-chapter(  
  "/02-main/03-elements.typ",  
  after: <sec:elem>,  
  before: <sec:ref>,  
)  
// or  
#add-chapter(  
  after: <sec:elem>,  
  before: <sec:ref>,  
) [  
  Content of the Chapter  
]
```

4.5 Images

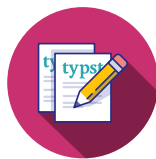
4.5.1 Alignment

left



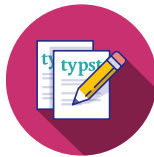
```
#image("/04-resources/icon.svg",
  width: 2cm)
```

center



```
#align(center,
  image("/04-resources/icon.svg",
    width: 2cm)
)
```

right



```
#align(right,
  image("/04-resources/icon.svg",
    width: 2cm)
)
```

4.5.2 Caption

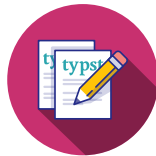


Abbildung 1 - One image one caption

```
#figure(
  image("/04-resources/icon.svg",
    width: 2cm),
  caption: [One image one caption]
) <fig:icon>
```

4.5.3 Cluster

Two images one caption



Tabelle 1 - Multiple images **one** caption

```
#figure(
  table(
    columns: 2,
```

```

stroke: none,
align: center + horizon,
image(icon, width: 2cm),image(icon, width: 2cm)
),
caption: [Multiple images *one* caption]
)

```

Four images one caption

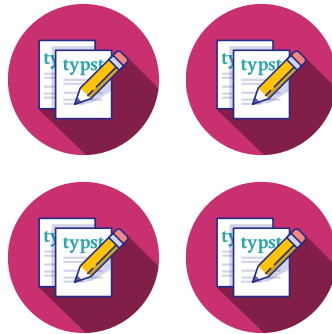


Tabelle 2 - Multiple images **one** caption

```

#figure(
  table(
    columns: 2,
    stroke: none,
    align: center + horizon,
    image(icon, width: 2cm), image(icon, width: 2cm),
    image(icon, width: 2cm), image(icon, width: 2cm),
  ),
  caption: [Multiple images *one* caption]
)

```

Two images two caption



Abbildung 2 - Caption left image Abbildung 3 - Caption right image

```

#align(center,
  table(
    columns: 2,
    stroke: none,
    align: center + horizon,
    figure(image(icon, width: 2cm), caption: [Caption left image]),
    figure(image(icon, width: 2cm), caption: [Caption right image]),
  ))

```

Four images four caption

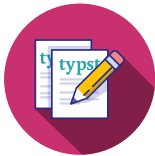


Abbildung 4 - Caption topleft image

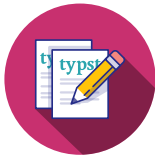


Abbildung 5 - Caption topright image

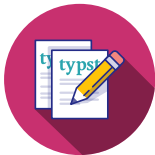


Abbildung 6 - Caption bottomleft image

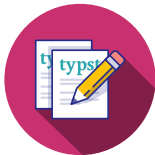


Abbildung 7 - Caption bottomright image

```
#align(center,
  table(
    columns: 2,
    stroke: none,
    align: center + horizon,
    figure(image(icon, width: 2cm), caption: [Caption topleft image]),
    figure(image(icon, width: 2cm), caption: [Caption topright image]),
    figure(image(icon, width: 2cm), caption: [Caption bottomleft image]),
    figure(image(icon, width: 2cm), caption: [Caption bottomright image]),
  ))
```

4.6 Tables

For new table use the integrated **#table** command for legacy the **tablex** plugin is also imported.

```
#import "@preview/tablex:0.0.9" : *
// or
#import "/00-templates/helpers.typ": *
```

Tables with and without caption

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

Tabelle 3 - Table caption

```
table(
  columns: 3,
  align: center + horizon,
  table.header([], [*Col1*] ,
[*Col2*]),),
[*Row1*], "cell-0-0", "cell-1-0",
```

```
figure(
  table(
    columns: 3,
    align: center + horizon,
    table.header([], [*Col1*] ,
[*Col2*]),),
[*Row1*], "cell-0-0", "cell-1-0",
```

```
[*Row2*], "cell-0-1", "cell-1-1",  
)
```

```
[*Row2*], "cell-0-1", "cell-1-1",  
) ,  
kind: table,  
caption: [Table Caption]  
)
```

Tables with cell spans

	Col1	Col2
Row1	cell-0	cell-1-0
Row2		cell-1-1

	Col1	Col2
Row1	cell-0	
Row2	cell-0-1	cell-1-1

```
table(  
  columns: 3,  
  align: center + horizon,  
  table.header([], [*Col1*] ,  
[*Col2*]),),  
  [*Row1*], table.cell(rowspan: 2)  
[cell-0], "cell-1-0",  
  [*Row2*],  
  "cell-1-1",  
)
```

```
table(  
  columns: 3,  
  align: center + horizon,  
  table.header([], [*Col1*] ,  
[*Col2*]),),  
  [*Row1*], table.cell(colspan: 2)  
[cell-0],  
  [*Row2*], "cell-0-1", "cell-1-1",  
)
```

Table Design

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

```
table(  
  columns: 3,  
  align: center + horizon,  
  table.header([], [*Col1*] ,  
[*Col2*]),),  
  [*Row1*], "cell-0-0", "cell-1-0",  
  [*Row2*], "cell-0-1", "cell-1-1",  
)
```

```
table(  
  columns: 3,  
  align: center + horizon,  
  table.header([], [*Col1*] ,  
[*Col2*]),),  
  [*Row1*], "cell-0-0", "cell-1-0",  
  [*Row2*], "cell-0-1", "cell-1-1",  
)
```

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

```
table(  
  columns: 3,
```

```
table(  
  columns: 3,
```

```
align: center + horizon,
  table.vline(x:0, stroke: none),
  table.vline(x:1 , stroke: blue),
  table.vline(x:2),
  table.header([],
    [*Col1*] , [*Col2*],
  table.hline(stroke: red)),
  [*Row1*], "cell-0-0", "cell-1-0",
  table.hline(),
  [*Row2*], "cell-0-1", "cell-1-1",
)
```

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

```
align: center + horizon,
stroke: (x:none),
table.header([], [*Col1*] ,
[*Col2*],),
[*Row1*], "cell-0-0", "cell-1-0",
[*Row2*], "cell-0-1", "cell-1-1",
),
```

	Col1	Col2
Row1	cell-0-0	cell-1-0
Row2	cell-0-1	cell-1-1

```
table(
  columns: 3,
  align: center + horizon,
  stroke: (y:none),
  table.header([], [*Col1*] ,
[*Col2*],),
  [*Row1*], "cell-0-0", "cell-1-0",
  [*Row2*], "cell-0-1", "cell-1-1",
),
```

```
table(
  columns: 3,
  align: center + horizon,
  stroke: (x:none),
  table.hline(y:0, stroke:none), //
remove first line
  table.hline(y:3, stroke:none), //
remove last line
  table.vline(x:1),
  table.vline(x:2, start:1, end:2,
stroke: red),
  table.header([], [*Col1*] ,
[*Col2*],),
  [*Row1*], "cell-0-0", "cell-1-0",
  [*Row2*], "cell-0-1", "cell-1-1",
),
```

```
#table(
  columns: 3,
  align: center + horizon,
  table.vline(x:0, stroke: none), table.vline(x:1 , stroke: blue), table.vline(x:2),
  table.header([], [*Col1*] , [*Col2*], table.hline(stroke: red)),
  [*Row1*], "cell-0-0", "cell-1-0", table.hline(),
  [*Row2*], "cell-0-1", "cell-1-1",
)
```

<i>c</i>	<i>b</i>	<i>a</i>	<i>cb</i>	<i>ba</i>	<i>y</i>
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	0	0
0	1	1	0	1	0

<i>c</i>	<i>b</i>	<i>a</i>	<i>cb</i>	<i>ba</i>	<i>y</i>
1	0	0	0	0	0
1	0	1	0	0	1
1	1	0	1	0	1
1	1	1	1	1	1

```
#table(  
  columns: 6,  
  stroke: none,  
  align: center+ horizon,  
  table.vline(x:1, stroke:0.5pt), table.vline(x:2, stroke:0.5pt), table.vline(x:3),  
  table.vline(x:4, stroke:0.5pt), table.vline(x:5, stroke:0.5pt),  
  table.header([$c$], [$b$], [$a$], [$c b$], [$b a$], [$y$], table.hline(stroke:  
1pt)),  
  [`0`], [`0`], [`0`], [`0`], [`0`], [`0`], table.hline(stroke: 0.5pt),  
  [`0`], [`0`], [`1`], [`0`], [`0`], [`1`], table.hline(stroke: 0.5pt),  
  [`0`], [`1`], [`0`], [`0`], [`0`], [`0`], table.hline(stroke: 0.5pt),  
  [`0`], [`1`], [`1`], [`0`], [`1`], [`0`], table.hline(stroke: 1pt),  
  [`1`], [`0`], [`0`], [`0`], [`0`], [`0`], table.hline(stroke: 0.5pt),  
  [`1`], [`0`], [`1`], [`0`], [`0`], [`1`], table.hline(stroke: 0.5pt),  
  [`1`], [`1`], [`0`], [`1`], [`0`], [`1`], table.hline(stroke: 0.5pt),  
  [`1`], [`1`], [`1`], [`1`], [`1`], [`1`],  
)
```

4.6.1 Karnaugh Tables

```
#import "/00-templates/karnaugh.typ": *
```

				<i>c</i>
				<i>b</i>
<i>y</i>				
	1	0	1	0
	1	0	1	1
				<i>a</i>

```
#karnaugh(content:((1, 0, 1, 0),  
  (1, 0, 1, 1),))
```

				<i>d</i>
				<i>c</i>
<i>y</i>				
	1	0	0	0
	1	0	0	1
	1	1	0	1
	1	0	1	1
				<i>a</i>
				<i>b</i>

```
#karnaugh(content:((1, 0, 0, 0),  
  (1, 0, 0, 1),  
  (1, 1, 0, 1),  
  (1, 0, 1, 1),))
```

$$\begin{array}{c} \overline{d} \\ \overline{c} \\ y \end{array} \begin{array}{|c|c|c|c|} \hline 1 & 0 & 0 & 0 \\ \hline 1 & 0 & 1 & 1 \\ \hline 1 & 0 & 1 & 1 \\ \hline 1 & 0 & 0 & 1 \\ \hline \end{array} \left| \begin{array}{c} a \\ b \end{array} \right.$$

$$\begin{array}{c} \overline{e} \\ \overline{d} \\ \overline{c} \\ y \end{array} \begin{array}{|c|c|c|c|} \hline 1 & 0 & 0 & 1 \\ \hline 1 & 0 & 0 & 1 \\ \hline 1 & 1 & 0 & 1 \\ \hline 1 & 1 & 0 & 1 \\ \hline \end{array} \left| \begin{array}{c} a \end{array} \right.$$

```
#karnaugh(content: ((1, 0, 0,
0),
(1, 0, 1,
1),
(1, 0, 1,
1),
(1, 0, 0,
1),
(1, 0, 0,
1),
(1, 0, 0,
1),
(1, 1, 0,
1),
(1, 1, 0,
1),,))
```

$$\begin{array}{c} \overline{Q_3} \\ \overline{Q_2} \\ Q_0^+ \end{array} \begin{array}{|c|c|c|c|} \hline 1 & 0 & 0 & 0 \\ \hline 1 & 0 & 1 & 1 \\ \hline 1 & 0 & 1 & 1 \\ \hline 1 & 0 & 0 & 1 \\ \hline \end{array} \left| \begin{array}{c} Q_0 \\ Q_1 \end{array} \right.$$

$$\begin{array}{c} \overline{Q_4} \\ \overline{Q_3} \\ \overline{Q_2} \\ Q_0^+ \end{array} \begin{array}{|c|c|c|c|} \hline 1 & 0 & 0 & 1 \\ \hline 1 & 0 & 0 & 1 \\ \hline 1 & 1 & 0 & 1 \\ \hline 1 & 1 & 0 & 1 \\ \hline \end{array} \left| \begin{array}{c} Q_0 \\ Q_1 \end{array} \right.$$

,

```
#karnaugh(inputs: ($Q_0$, $Q_1$, $Q_2$, $Q_3$, $Q_4$),
output: [$Q_0^+$],
content: ((1, 0, 0, 0),
(1, 0, 1, 1),
(1, 0, 1, 1),
(1, 0, 0, 1),

(1, 0, 0, 1),
(1, 0, 0, 1),
(1, 1, 0, 1),
(1, 1, 0, 1),,))
```

4.7 Icon Boxes

```
#import "/00-templates/boxes.typ": *
```



```
#infobox()["infobox"]
```



```
#ideabox()["ideabox"]
```



```
#warningbox()["warningbox"]
```



```
#importantbox()["importantbox"]
```



```
#firebox()["firebox"]
```



```
#rocketbox()["rocketbox"]
```



```
#todobox()["todobox"]
```



```
#iconbox(icon: "/04-resources/placeholder.svg", linecolor:  
hei-blue)["iconbox"]
```

```
#iconbox(linecolor: hei-pink)["iconbox without icon"]
```

4.8 Color Boxes

```
#import "/00-templates/boxes.typ": *
```

Exercise

Some text

```
#colorbox(title: "Exercise", color: hei-blue)[Some text]
```

Attention

Some text

```
#colorbox(title: "Attention", color: hei-pink)[Some text]
```

Consider

Some text

```
#slanted-colorbox(title: "Consider", color: hei-green)[Some text]
```

Information

Some text

```
#slanted-colorbox(title: "Information", color: hei-orange)[Some text]
```

4.8.1 Todo Box

```
#import "/00-templates/boxes.typ": *
```

TODO

This is not finished

```
#todo("This is not finished")
```

4.8.2 Option Style

The option style allows to underlight a text depending on the type or state of the document. Within the `/01-settings/metadata.typ` the `option.type` can be set to **draft** or **final** or other types.

```
#import "/00-templates/boxes.typ": *
```

```
#option-style(type:"draft")["This text has an option style and is shown in the case  
the type is draft"]
```

```
#option-style(type:"final")["This text has an option style and is shown in the case  
the type is final"]
```

```
#option-style(type:"minimal")["This text has an option style and is shown in the  
case the type is minimal"]
```

„This text has an option style and is shown in the case the type is full“

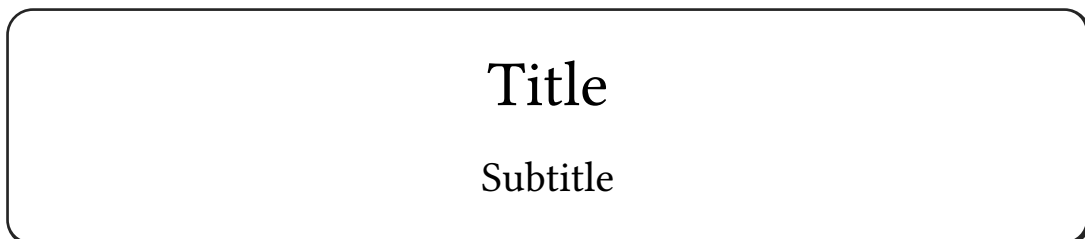
```
#option-style(type:"full")["This text has an option style and is shown in the case  
the type is full"]
```

```
#option-style(type:"student")["This text has an option style and is shown in the  
case the type is student"]
```

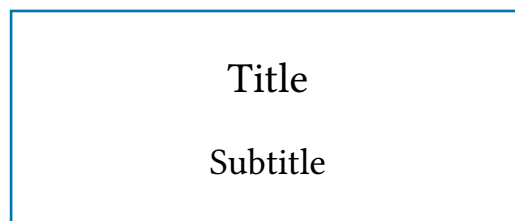
```
#option-style(type:"solution")["This text has an option style and is shown in the  
case the type is solution"]
```

4.9 Title Box

```
#import "/00-templates/sections.typ": *
```



```
#titlebox(title: [Title], subtitle: [Subtitle])
```



```
#titlebox(width: 50%, radius: 0pt, border: 1pt, linecolor: hei-blue, titlesize: larger, subtitlesize: large, title: [Title], subtitle: [Subtitle])
```

Title

```
#titlebox(linecolor: hei-green, titlesize: larger, subtitlesize: large, title: [Title])
```

4.10 Exam Header

```
#import "/00-templates/sections.typ": *
```

Name: -----

```
#exam-header(nbr-ex: 0, lang: "en")
```

Name: -----

Grade

```
#exam-header(nbr-ex: 1, lang: "en")
```

Name: -----

1Grade

(10)

```
#exam-header(nbr-ex: 2, pts: 10, lang: "en")
```

Name: -----

12Grade

(10)

(10)

```
#exam-header(nbr-ex: 3, pts: 10, lang: "en")
```

123Grade

Name:

(10)	(10)	(10)	

```
#exam-header(nbr-ex: 4, pts: 10, lang: "en")
```

Name:

1	2	3	4	Grade
(10)	(10)	(10)	(10)	

```
#exam-header(nbr-ex: 5, pts: 10, lang: "en")
```

Name:

1	2	3	4	5	Grade
(10)	(10)	(10)	(10)	(10)	

```
#exam-header(nbr-ex: 6, pts: 10, lang: "en")
```

Name:

1	2	3	4	5	6	Grade
(10)	(10)	(10)	(10)	(10)	(10)	

```
#exam-header(nbr-ex: 7, pts: 10, lang: "en")
```

Name:

1	2	3	4	5	6	7	Grade
(10)	(10)	(10)	(10)	(10)	(10)	(10)	

```
#exam-header(nbr-ex: 8, pts: 10, lang: "en")
```

Name:

1	2	3	4	5	6	7	8	Grade
(10)	(10)	(10)	(10)	(10)	(10)	(10)	(10)	

```
#exam-header(nbr-ex: 9, pts: 10, lang: "en")
```

1	2	3	4	5	6	7	8	9	Grade
---	---	---	---	---	---	---	---	---	-------

Name:

(10)	(10)	(10)	(10)	(10)	(10)	(10)	(10)	(10)	(10)

```
#exam-header(nbr-ex: 10, pts: 10, lang: "en")
```

4.11 Exam Reminder

```
#import "/00-templates/sections.typ": *
```



Exam Reminder:

You can only use the following items:

- a laptop without internet connection
- a pocketcalculator
- all paper documents you want

It is forbidden to use generative AI.

Good Luck!

```
#exam-reminder-did(lang: "en")
```



Prüfungserinnerung:

Sie können nur die folgenden Gegenstände verwenden:

- ein Laptop ohne Internetanschluss
- einen Taschenrechner
- alle Papierdokumente

Es ist verboten, generative KI zu verwenden.

Viel Glück!

```
#exam-reminder-did(lang: "de")
```

Rappel d'examen :

Vous ne pouvez utiliser que les éléments suivants :



- un ordinateur portable sans connexion internet
- une calculatrice de poche
- tous les documents papier que vous souhaitez

Il est interdit d'utiliser l'IA générative.

Bonne chance!

```
#exam-reminder-did(lang: "fr")
```

Exam Reminder:

You can only use the following items:



- the two-page summary you created.
- a pocketcalculator

In addition, properly comment all high-level and assembler code to explain its purpose and how it fits into the program structure.

Good Luck!

```
#exam-reminder-car(lang: "en")
```

Prüfungserinnerung:

Sie können nur die folgenden Elemente verwenden:



- die zweiseitige Zusammenfassung, die Sie erstellt haben.
- einen Taschenrechner

Kommentieren Sie ausserdem den gesamten High-Level- und Assembler-Code ordnungsgemäss aus, um seinen Zweck und seine Einbindung in die Programmstruktur zu erklären.

Viel Glück!

```
#exam-reminder-car(lang: "de")
```

Rappel d'examen :

Vous ne pouvez utiliser que les éléments suivants :

- le résumé de deux pages que vous avez créé.
- une calculatrice de poche

Commenter également tout le code de haut niveau et le code assembleur de manière appropriée afin d'expliquer son but et son intégration dans la structure du programme.

Bonne chance!

```
#exam-reminder-car(lang: "fr")
```

4.12 Exercise Message

```
#import "/00-templates/sections.typ": *
```

**Solution vs. Hints:**

While not every response provided herein constitutes a comprehensive solution, some serve as helpful hints intended to guide you toward discovering the solution independently. In certain instances, only a portion of the solution is presented.

```
#exercises-solution-hints(lang: "en")
```

**Lösung vs. Hinweise:**

Nicht alle hier gegebenen Antworten sind vollständige Lösungen. Einige dienen lediglich als Hinweise, um Ihnen bei der eigenständigen Lösungsfindung zu helfen. In anderen Fällen wird nur ein Teil der Lösung präsentiert.

```
#exercises-solution-hints(lang: "de")
```

**Solution vs. Hints:**

Toutes les réponses fournies ici ne sont pas des solutions complètes. Certaines ne sont que des indices pour vous aider à trouver la solution vous-même. Dans d'autres cas, seule une partie de la solution est fournie.

```
#exercises-solution-hints(lang: "fr")
```

4.13 Word/Character Count

With the plugin wordometer is possible to count the words and characters of a text.

```
#import "@preview/wordometer:0.1.4": word-count
```

The following section contains 50 Words and 295 Characters without counting this text.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat voluptatem. Ut enim aequaleam animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

```
#word-count(total => [
  #[The following section contains #total.words Words and #total.characters
  Characters without counting this text.]<no-wc>
  #lorem(50)
], exclude: <no-wc>)
```



The character count does not include spaces.

5 | References

5.1 Links

Example	Raw
https://tschinz.github.io	<code>https://tschinz.github.io</code>
https://tschinz.github.io	<code>#link("https://tschinz.github.io")</code>
See https://tschinz.github.io	<code>#link("https://tschinz.github.io") [See https://tschinz.github.io]</code>
silvan.zahno@hevs.ch	<code>#link("mailto:silvan.zahno@hevs.com") [silvan.zahno@hevs.ch]</code>
	<code>#link("https://tschinz.github.io") [#image(icon, width:0.5cm)]</code>

Tabelle 8 - Links

5.2 Crossreferences

In the document the following references were added.

```
= References <sec:ref>
== Links <sec:links>
#figure(image("/04-resources/icon.svg", width: 2cm)) <fig:icon>
#figure(table(...), kind:table) <tab:links>
#figure(align(left, raw(...)) <code-ref>
$ \sum_{k=1}^n k = (n(n+1)) / 2 $ <math-eq1> #ref(<math-eq1>)
```

Listing 1 - Label inserts

They can be references as follows:

Type	Example	Raw
Section	Abschnitt 5	<code>@sec:ref</code>
Subsection	Abschnitt 5.1	<code>@sec:links</code>
Figure	Abbildung 1	<code>@fig:icon</code>
Table	Tabelle 8	<code>@tab:links</code>
Code	Listing 1	<code>@code-ref</code>
Equation	Gleichung 1	<code>@math-eq1</code>

5.3 External References

Example

[1]

[1, p.7ff]

[1]

Raw

```
#cite(label("zahnoDynamicProjectPlanning2023"))
```

```
#cite(<zahnoDynamicProjectPlanning2023>,
supplement:[p.7ff])
```

```
@zahnoDynamicProjectPlanning2023
```

5.4 Glossary

The glossary entries need to be defined in `/03-tail/glossary.typ`.

```
#let entry-list = (
  (
    key   : "hei",
    short : "HEI",
    long  : "Haute École d'Ingénierie",
  ),
  (
    key       : "fpga",
    short     : "FPGA",
    plural    : "FPGAs",
    long      : "Field Programmable Gate Array",
    longplural : "Field Programmable Gate Arrays",
    group     : "Technology"
    description : "A field-programmable gate array (FPGA) is an integrated circuit
designed to be configured by a customer or a designer after manufacturing – hence
the term 'field-programmable'.",
  ),
)
```

For the glossary functions the „import“ of `/00-templates/helpers.typ` is needed. Underneath it uses the glossarium plugin. It can also be used for acronyms.

```
#import "/00-templates/helpers.typ": *
```

Example

Raw

Haute École d'Ingénierie (HEI) and a second time HEI

```
#gls("hei") and a second time
#gls("hei")
```

Haute École d'Ingénierie (HEI)

```
#gls("hei", long:true)
```

whatever you want

```
#gls("hei", display: "whatever you
want"))
```

Field Programmable Gate Arrays (FPGAs) and a second time FPGAs

```
#glspl("fpga") and a second time
#glspl("fpga")
```

Field Programmable Gate Arrays (FPGAs)

```
#glspl("fpga", long: true))
```

Infotronics (IT) is a specialization of Systems Engineering (SYND) which is part of the HEI. The second time a glossary entry is used the short form will be used: see IT and SYND. To get the long form back use Systems Engineering (SYND).

6 | Code

inline monospaced string

```
`inline monospaced string`
```

```
fn main() {println!("Hello world!")}
```

```
raw(lang:"rust",
  "fn main() {println!(\"Hello world!
  \")"
)
```

```
-- Test 2: INPUT sX, pp
opCode <= "INPUT sX, pp    ";
code    <= "00010";
cIn     <= '0';
A       <= "11110000";
B       <= "00001111";
wait for clockPeriod;
assert Y = "00001111"
  report "test 2 INPUT wrong"
  severity note
```

```
raw(block:true, lang:"vhdl",
  read("code-example.vhdl"))
)
```

```
fn main() {
  println!("Hello world!")
}
```

```
```\rust
fn main() {
 println!("Hello world!")
}
```\
```

```
fn main() {
  println!("Hello world!")
}
```

Listing 2 - Rust Code

```
#figure(
  align(left,
    ```\rust
 fn main() {
 println!("Hello world!")
 }
    ```\
  ),
  caption: [Rust Code],
)
```

A plugin allows to get linenumbers

```
#import "@preview/codelst:2.0.2": sourcecode
```

```
1 fn main() {
2   println!("Hello world!")
3 }
```

```
#sourcecode()[
  ```\rust
 fn main() {
```

```
println!("Hello world!")
}
```\]
```

7 | Math Equations

Inline math

Let a and b , and c be the side of a right-angled triangle.

Let a and b , and c be the side of a right-angled triangle.

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n k = (n(n+1)) / 2,$$

Fullline math

$$a^2 + b^2 = c^2 \tag{1}$$

$a^2 + b^2 = c^2$
 $\tag{1}$

Math with caption

$$\sum_{k=1}^n k = \frac{n(n+1)}{2} \tag{2}$$

Gleichung 8 - Some proof

```
#figure(
  $ sum_(k=1)^n k = (n(n+1)) / 2 $,
  caption: [Some proof]
)
```

7.1 Align

Formula	Raw
$\begin{aligned} a_1 &= b_1 + c_1 = z_1 \\ a_2 &= b_2 + c_2 - d_2 + e_2 = z_1 \end{aligned} \tag{3}$	<pre>\$ a_1 = b_1 + c_1 = z_1 + \ a_2 = b_2 + c_2 - d_2 + e_2 = z_1 \$</pre>
$\begin{aligned} a_1 &= b_1 + c_1 &= z_1 \\ a_2 &= b_2 + c_2 - d_2 + e_2 = z_1 \end{aligned} \tag{4}$	<pre>\$ a_1 &= b_1 + c_1 &= z_1 \ a_2 &= b_2 + c_2 - d_2 + e_2 &= z_1 \$</pre>

7.2 Symbols

This is an incomplete list for all symbols goto [here](#)
 Outside of the $\$$ math environment the symboils can be accessed with .

7.2.1 Accents

Sym- bol	Raw	Sym- bol	Raw	Sym- bol	Raw
$\hat{x}$	<code>\$grave(x)\$</code>	$\acute{x}$	<code>\$acute(x)\$</code>	$\hat{x}$	<code>\$hat(x)\$</code>
$\tilde{x}$	<code>\$tilde(x)\$</code>	$\breve{x}$	<code>\$breve(x)\$</code>	$\dot{x}$	<code>\$dot(x)\$</code>
$\ddot{x}$	<code>\$dot.double(x)\$</code>	$\ddot{x}$	<code>\$dot.triple(x)\$</code>	$\ddot{x}$	<code>\$dot.quad(x)\$</code>
$\ddot{x}$	<code>\$diaer(x)\$</code>	$\mathring{x}$	<code>\$circle(x)\$</code>	$\ddot{x}$	<code>\$acute.double(x)\$</code>
$\tilde{x}$	<code>\$caron(x)\$</code>	$\vec{x}$	<code>\$arrow(x)\$</code>	$\vec{x}$	<code>\$arrow.l(x)\$</code>
$\cancel{x}$	<code>\$cancel(x)\$</code>	$\bar{x}$	<code>\$macron(x)\$</code>	$\overline{xyz}$	<code>\$overline(xyz)\$</code>
$\overline{xyz}$	<code>\$overline(xyz)\$</code>	$\underbrace{xyz}$	<code>\$underbrace(xyz)\$</code>	$\overbrace{xyz}$	<code>\$overbrace(xyz)\$</code>
$\underbrace{xyz}$	<code>\$underbracket(xyz)\$</code>	$\overbrace{xyz}$	<code>\$overbracket(xyz)\$</code>	$\overbrace{xyz}$	<code>\$overbracket(xyz)\$</code>

7.2.2 Equals & Operators

Sym- bol	Raw	Sym- bol	Raw	Sym- bol	Raw
$=$	<code>\$=\$</code>	$=$	<code>\$eq\$</code>	$\neq$	<code>\$eq.not\$</code>
$\neq$	<code>\$!= \$</code>	$\equiv$	<code>\$equiv\$</code>	$\neq$	<code>\$equiv.not\$</code>
$\simeq$	<code>\$tilde.eq\$</code>	$\not\equiv$	<code>\$tilde.eq.not\$</code>	$=$	<code>\$eq.small\$</code>
$\geq$	<code>\$gt.eq\$</code>	$\nless$	<code>\$gt.eq.not\$</code>	$\leq$	<code>\$lt.eq\$</code>
$\nless$	<code>\$lt.eq.not\$</code>	$\approx$	<code>\$approx\$</code>	$\approx$	<code>\$approx.eq\$</code>
$\napprox$	<code>\$approx.not\$</code>	$:$	<code>\$colon\$</code>	$:=$	<code>\$colon.eq\$</code>
$=:$	<code>\$eq.colon\$</code>	$::=$	<code>\$colon.double.eq\$</code>	$+$	<code>\$+\$</code>
$+$	<code>\$plus\$</code>	$+$	<code>\$plus.small\$</code>	$\pm$	<code>\$plus.minus\$</code>
$\oplus$	<code>\$plus.circle\$</code>	$-$	<code>\$-\$</code>	$-$	<code>\$minus\$</code>
$\mp$	<code>\$minus.plus\$</code>	$\ominus$	<code>\$minus.circle\$</code>		

7.2.3 Scripts

Sym- bol	Raw	Sym- bol	Raw	Sym- bol	Raw
x_1	<code>\$x_1\$</code>	x_{12}	<code>\$x_(12)\$</code>	x_1	<code>\$scripts(x)_1\$</code>
x_1	<code>\$x_1\$</code>	x_{12}	<code>\$x_(12)\$</code>	x_1	<code>\$scripts(x)_1\$</code>
x_1^2	<code>\$x_1^2\$</code>	x_{12}^{34}	<code>\$x_(12)^(34)\$</code>	x_1^2	<code>\$scripts(x)_1^2\$</code>
x_1^2	<code>\$x_1^2\$</code>	x_{12}^{34}	<code>\$x_(12)^(34)\$</code>	x_1^2	<code>\$scripts(x)_1^2\$</code>

7.2.4 Special Elements

Symbol	Raw	Symbol	Raw
--------	-----	--------	-----

$$\binom{n}{k} \quad (5) \quad \$ \text{binom}(n, k) \$$$

$$\left\lfloor \frac{1}{2} \right\rfloor \quad (7) \quad \$ \text{round}(1, 2) \$$$

$$\begin{pmatrix} 1 & 2 & \dots & 10 \\ 2 & 2 & \dots & 10 \\ \vdots & \vdots & \ddots & \vdots \\ 10 & 10 & \dots & 10 \end{pmatrix} (9)$$

```
$ mat(
  1, 2, ...,
  10;
  2, 2, ...,
  10;
  dots.v,
  dots.v,
  dots.down,
  dots.v;
  10, 10, ...,
  10;
)
```

$$\sum_{k=0}^n a_k \quad (11) \quad \$ \text{sum_}(k=0)^n \text{ a_k} \$$$

$$\sqrt[3]{x} \quad (13) \quad \$ \text{root}(3, x) \$$$

```
$ f(x, y) := cases(
  1 "if" (x dot y)/2 <= 0,
  2 "if" x "is even",
  3 "if" x in NN,
  4 "else",
)
```

$$\frac{1}{2} \quad (16) \quad \$ \text{frac}(1,2) \$$$

$$\frac{(x+1)}{(x+2)} \quad (18) \quad \$ ((x+1))/(x+2) \$$$

$$n! = \prod_{k=1}^n k \quad (20) \quad \$ \text{product_}(k=1)^n \text{ k} \$$$

$$\int \quad (22) \quad \$ \text{integral} \$$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad (6) \quad \$ \text{vec}(1, 2, \text{delim: "["]}) \$$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad (8) \quad \$ \text{mat}(1,2; 3,4) \$$$

$$\sum a_k \quad (10) \quad \$ \text{sum} \text{ a_k} \$$$

$$\sum_{k=0}^n a_k \quad (12) \quad \$ \text{scripts}(\text{sum_}(k=0)^n \text{ a_k} \$$$

$$f(x, y) := \begin{cases} 1 & \text{if } \frac{x \cdot y}{2} \leq 0 \\ 2 & \text{if } x \text{ is even} \\ 3 & \text{if } x \in \mathbb{N} \\ 4 & \text{else} \end{cases} \quad (14)$$

$$\frac{1}{2} \quad (15) \quad \$ 1/2 \$$$

$$\frac{x+1}{x+2} \quad (17) \quad \$ (x+1)/(x+2) \$$$

$$\prod \quad (19) \quad \$ \text{product} \$$$

$$n! = \prod_{k=1}^n k \quad (21) \quad \$ \text{n!} = \text{scripts}(\text{product_}(k=1)^n \text{ k} \$$$

$$\int_a^b f(x) \quad (23) \quad \$ \text{integral} \$$$

7.2.5 Alphabeth

Symbol

Raw

$\alpha\beta\gamma\delta\epsilon\zeta\eta\theta\iota\kappa\lambda\mu\nu\xi\omicron\rho\sigma\tau\upsilon\varphi\chi\psi\omega$	<code>\$alpha beta gamma delta epsilon zeta eta theta iota kappa lambda mu nu xi omicron pi rho sigma tau upsilon phi chi psi omega\$</code>
$\text{ΑΒΓΔΕΖΗΘΙΚΑΜΝΞΟΠΡΣΤΥΦΧΨΩ}$	<code>\$Alpha Beta Gamma Delta Epsilon Zeta Eta Theta Iota Kappa Lambda Mu Nu Xi Omicron Pi Rho Sigma Tau Upsilon Phi Chi Psi Omega\$</code>
$\text{ABCDEFGHIJKLMNOPQRSTUVWXYZ}$	<code>\$AA BB CC DD EE FF GG HH II JJ KK LL MM NN OO PP QQ RR SS TT UU VV WW XX YY ZZ\$</code>

7.2.6 Logical

Symbol	Raw	Symbol	Raw	Symbol	Raw
$\wedge$	<code>\$and\$</code>	$\bigwedge$	<code>\$and.big\$</code>	$\&$	<code>\$amp\$</code>
$\vee$	<code>\$or\$</code>	$ $	<code>\$bar.v\$</code>	$*$	<code>\$ast.op\$</code>
$*$	<code>\$ast.basic\$</code>	$*$	<code>\$ast.low\$</code>	$\oplus$	<code>\$plus.circle\$</code>
$\oplus$	<code>\$plus.circle.big\$</code>				

7.2.7 Operators

Sym- bol	Raw	Sym- bol	Raw	Sym- bol	Raw
$\sin x$	<code>\$sin x\$</code>	$\cos x$	<code>\$cos x\$</code>	$\tan x$	<code>\$tan x\$</code>
$\arcsin x$	<code>\$arcsin x\$</code>	$\arccos x$	<code>\$arccos x\$</code>	$\arctan x$	<code>\$arctan x\$</code>
$\sinh x$	<code>\$sinh x\$</code>	$\cosh x$	<code>\$cosh x\$</code>	$\tanh x$	<code>\$tanh x\$</code>
$\arg x$	<code>\$arg x\$</code>	$\csc x$	<code>\$csc x\$</code>	$\deg x$	<code>\$deg x\$</code>
$\det x$	<code>\$det x\$</code>	$\dim x$	<code>\$dim x\$</code>	$\exp x$	<code>\$exp x\$</code>
$\bmod x$	<code>\$mod x\$</code>	$\inf x$	<code>\$inf x\$</code>	$\log x$	<code>\$log x\$</code>
$\lim x$	<code>\$lim x\$</code>	$\liminf x$	<code>\$liminf x\$</code>	$\limsup x$	<code>\$limsup x\$</code>
$\min x$	<code>\$min x\$</code>	$\max x$	<code>\$max x\$</code>	$\sup x$	<code>\$sup x\$</code>

7.2.8 Arrows

SymRaw	SymRaw	SymRaw	
	Arrows right		
$\rightarrow$	<code>\$arrow\$</code>	$\mapsto$	<code>\$arrow.bar\$</code>
$\mapsto$	<code>\$arrow.bar.long\$</code>	$\Rightarrow$	<code>\$arrow.double.long\$</code>
$\mapsto$	<code>\$arrow.double.bar\$</code>	$\Rightarrow$	<code>\$arrow.double.bar.long\$</code>
$\Rightarrow$	<code>\$arrow.stroked\$</code>	$\Rightarrow$	<code>\$arrow.quad\$</code>
$\hookrightarrow$	<code>\$arrow.filled\$</code>	$\dashrightarrow$	<code>\$arrow.dashed\$</code>
$\hookrightarrow$	<code>\$arrow.curve\$</code>	$\looparrowright$	<code>\$arrow.loop\$</code>
	Arrows left		

	<code>\$arrow.l\$</code>		<code>\$arrow.l.long\$</code>		<code>\$arrow.l.bar\$</code>
	<code>\$arrow.l.bar.long\$</code>		<code>\$arrow.l.double\$</code>		<code>\$arrow.l.double.long\$</code>
	<code>\$arrow.l.double.bar\$</code>		<code>\$arrow.l.double.bar.long\$</code>		<code>\$arrow.l.quad\$</code>
	<code>\$arrow.l.stroked\$</code>		<code>\$arrow.l.filled\$</code>		<code>\$arrow.l.dashed\$</code>
	<code>\$arrow.l.curve\$</code>		<code>\$arrow.l.squiggly\$</code>		<code>\$arrow.l.loop\$</code>
Double Arrows Left Right					
	<code>\$arrow.l.r\$</code>		<code>\$arrow.l.r.not\$</code>		<code>\$arrow.l.r.long\$</code>
	<code>\$arrow.l.r.double\$</code>		<code>\$arrow.l.r.double.long\$</code>		<code>\$arrow.l.r.double.not\$</code>
	<code>\$arrow.l.r.stroked\$</code>		<code>\$arrow.l.r.filled\$</code>		<code>\$arrow.l.r.wave\$</code>
Arrows Top					
	<code>\$arrow.t\$</code>		<code>\$arrow.t.bar\$</code>		<code>\$arrow.t.double\$</code>
	<code>\$arrow.t.triple\$</code>		<code>\$arrow.t.quad\$</code>		<code>\$arrow.t.stroked\$</code>
	<code>\$arrow.t.filled\$</code>		<code>\$arrow.t.dashed\$</code>		<code>\$arrow.t.curve\$</code>
Arrows Bottom					
	<code>\$arrow.b\$</code>		<code>\$arrow.b.bar\$</code>		<code>\$arrow.b.double\$</code>
	<code>\$arrow.b.triple\$</code>		<code>\$arrow.b.quad\$</code>		<code>\$arrow.b.stroked\$</code>
	<code>\$arrow.b.filled\$</code>		<code>\$arrow.b.dashed\$</code>		<code>\$arrow.b.curve\$</code>
Double Arrows Top Bottom					
	<code>\$arrow.t.b\$</code>		<code>\$arrow.t.b.double\$</code>		<code>\$arrow.t.b.stroked\$</code>
	<code>\$arrow.t.b.filled\$</code>				
Arrows Diagonal Top Right					
	<code>\$arrow.tr\$</code>		<code>\$arrow.tr.double\$</code>		<code>\$arrow.tr.stroked\$</code>
	<code>\$arrow.tr.filled\$</code>		<code>\$arrow.tr.hook\$</code>		
Arrows Diagonal Bottom Right					
	<code>\$arrow.br\$</code>		<code>\$arrow.br.double\$</code>		<code>\$arrow.br.stroked\$</code>
	<code>\$arrow.br.filled\$</code>		<code>\$arrow.br.hook\$</code>		
Arrows Diagonal Bottom Left					
	<code>\$arrow.bl\$</code>		<code>\$arrow.bl.double\$</code>		<code>\$arrow.bl.stroked\$</code>
	<code>\$arrow.bl.filled\$</code>		<code>\$arrow.bl.hook\$</code>		
Arrows Diagonal Top Left					
	<code>\$arrow.tl\$</code>		<code>\$arrow.tl.double\$</code>		<code>\$arrow.tl.stroked\$</code>
	<code>\$arrow.tl.filled\$</code>		<code>\$arrow.tl.hook\$</code>		
Double Arrows Diagonal					
	<code>\$arrow.tl.br\$</code>		<code>\$arrow.tr.bl\$</code>		
Other Arrows					
	<code>\$arrow.cw\$</code>		<code>\$arrow.cw.half\$</code>		<code>\$arrow.ccw\$</code>

 `$arrow.ccw.half$`

7.2.9 Angles

Sym- bol	Raw	Sym- bol	Raw	Sym- bol	Raw
	<code>\$angle\$</code>		<code>\$angle.rev\$</code>		<code>\$angle.acute\$</code>
	<code>\$angle.acute\$</code>		<code>\$angle.arc\$</code>		<code>\$angle.arc.rev\$</code>
	<code>\$angle.l\$</code>		<code>\$angle.r\$</code>		<code>\$angle.l.double\$</code>
	<code>\$angle.r.double\$</code>		<code>\$angle.right\$</code>		<code>\$angle.right.rev\$</code>
	<code>\$angle.right.arc\$</code>		<code>\$angle.right.dot\$</code>		<code>\$angle.right.sq\$</code>
	<code>\$angle.spheric\$</code>		<code>\$angle.spheric.rev\$</code>		<code>\$angle.spheric.top\$</code>

7.2.10 Cool Symbols

Sym- bol	Raw	Sym- bol	Raw	Sym- bol	Raw
@	<code>\$at\$</code>	‰	<code>\$co\$</code>	©	<code>\$copyright\$</code>
Ⓒ	<code>\$copyright.sound\$</code>	°C	<code>\$degree C\$</code>	€	<code>\$euro\$</code>
\$	<code>\$dollar\$</code>	£	<code>\$pound\$</code>	₩	<code>\$won\$</code>
¥	<code>\$yen\$</code>	฿	<code>\$bitcoin\$</code>	°F	<code>\$degree F\$</code>
!	<code>\$excl\$</code>	¡	<code>\$excl.inv\$</code>	!!	<code>\$excl.double\$</code>
!?	<code>\$excl.quest\$</code>	↯	<code>\$arrow.zigzag\$</code>	⊗	<code>\$ast.circle\$</code>
⚙	<code>\$ast.triple\$</code>	χ	<code>\$chi\$</code>	☒	<code>\$floral\$</code>
⚔	<code>\$maltese\$</code>	¶	<code>\$pilcrow\$</code>	ℏ	<code>\$planck\$</code>
♣	<code>\$suit.club\$</code>	♦	<code>\$suit.diamond\$</code>	♥	<code>\$suit.heart\$</code>
♠	<code>\$suit.spade\$</code>	⚠	<code>\$triangle.stroked.nested2\$</code>		

7.2.11 Style

Symbol	Raw	Symbol	Raw
ABC123	<code>\$sans(A B C 1 2 3)\$</code>	ℳ℔℥123	<code>\$frak(A B C 1 2 3)\$</code>
ABC123	<code>\$mono(A B C 1 2 3)\$</code>	ABC123	<code>\$bb(A B C 1 2 3)\$</code>
ABC123	<code>\$cal(A B C 1 2 3)\$</code>		

Symbol Raw

$\sum_{i \in \mathbb{N}} 1 + i$

```
#show math.equation: set text(font: "Fira Math")
$sum_(i in NN) 1 + i$,
```

8 | Emoji Symbols

This is an incomplete list for all emoji goto [here](#)

If the emoji module is imported the `#emoji` can be removed

```
#import emoji: *
```

Sym Raw	Sym Raw	Sym Raw
	Smileys	
#emoji.face	#emoji.baby	#emoji.alien
#emoji.alien.monster	#emoji.alien.monster	#emoji.eye
#emoji.eyes	#emoji.face.grin	#emoji.face.angry
#emoji.face.angry.red	#emoji.face.anguish	#emoji.face.astonish
#emoji.face.bandage	#emoji.face.beam	#emoji.face.concern
#emoji.face.cool	#emoji.face.cover	#emoji.face.down
#emoji.face.down.sweat	#emoji.face.drool	#emoji.face.explode
#emoji.face.eyeroll	#emoji.face.friendly	#emoji.face.fear
#emoji.face.fear.sweat	#emoji.face.fever	#emoji.face.flush
#emoji.face.halo	#emoji.face.happy	#emoji.face.heart
#emoji.face.hearts	#emoji.face.heat	#emoji.face.kiss
#emoji.face.kiss.heart	#emoji.face.kiss.blush	#emoji.face.joy
#emoji.face.lie	#emoji.face.mask	#emoji.face.meh
#emoji.face.melt	#emoji.face.neutral	#emoji.face.party
#emoji.face.peek	#emoji.face.plead	#emoji.face.relief
#emoji.face.vomit	#emoji.face.unhappy	#emoji.face.wow
#emoji.face.yawn	#emoji.face.wink	#emoji.face.zip
#emoji.finger.l	#emoji.finger.t	#emoji.finger.b
#emoji.finger.m	#emoji.finger.front	#emoji.fingers.cross
#emoji.fingers.pinch	#emoji.fingers.snap	#emoji.fist.front
#emoji.fist.r	#emoji.fist.l	#emoji.fist.raised
#emoji.glasses	#emoji.glasses.sun	#emoji.hand.raised
#emoji.hand.r	#emoji.hand.l	#emoji.hand.t

Sym Raw

 #emoji.hand.b
 #emoji.hand.love
 #emoji.hand.pinch
 #emoji.hand.write
 #emoji.hands.heart

Sym Raw

 #emoji.hand.ok
 #emoji.hand.part
 #emoji.hand.rock
 #emoji.hands.folded
 #emoji.hands.shake

Sym Raw

 #emoji.hand.call
 #emoji.hand.peace
 #emoji.hand.wave
 #emoji.hands.clap
 #emoji.hand.splay

Drinks

Sym Raw

 #emoji.beer
 #emoji.coffee
 #emoji.glass.clink
 #emoji.teacup

Sym Raw

 #emoji.beer.clink
 #emoji.cup.straw
 #emoji.glass.milk
 #emoji.mate

Sym Raw

 #emoji.cocktail.tropical
 #emoji.glass.tumbler
 #emoji.teapot
 #emoji.soup

Animals

Sym Raw

 #emoji.ant
 #emoji.beetle
 #emoji.beetle.lady
 #emoji.bison
 #emoji.butterfly
 #emoji.cat
 #emoji.chipmunk
 #emoji.crocodile
 #emoji.dodo
 #emoji.duck
 #emoji.fairy
 #emoji.flamingo
 #emoji.giraffe

Sym Raw

 #emoji.badger
 #emoji.bee
 #emoji.beaver
 #emoji.boar
 #emoji.camel
 #emoji.chicken
 #emoji.cockroach
 #emoji.dino.pod
 #emoji.dog
 #emoji.egg
 #emoji.feather
 #emoji.fly
 #emoji.hedgehog

Sym Raw

 #emoji.bat
 #emoji.beetle
 #emoji.bear
 #emoji.bug
 #emoji.camel.dromedar
 #emoji.chicken.male
 #emoji.cricket
 #emoji.dino.rex
 #emoji.dolphin
 #emoji.elephant
 #emoji.fish
 #emoji.goat
 #emoji.hippo

Flowers

Sym Raw

 #emoji.fleur
 #emoji.flower.pink

Sym Raw

 #emoji.flower.hibiscus
 #emoji.flower.rose

Sym Raw

 #emoji.flower.lotus
 #emoji.flower.sun

Sym Raw

 #emoji.flower.wilted

Sym Raw

 #emoji.flower.yellow

Sym Raw

 #emoji.plant

Food

Sym Raw

 #emoji.avocado #emoji.broccoli #emoji.carrot #emoji.cucumber

Sym Raw

 #emoji.aubergine #emoji.burger #emoji.chocolate #emoji.fries

Sym Raw

 #emoji.banana #emoji.cake.slice #emoji.corn #emoji.honey

Objects

Sym Raw

 #emoji.bin #emoji.books #emoji.bell #emoji.broom #emoji.cabinet.file #emoji.compass

Sym Raw

 #emoji.bomb #emoji.book #emoji.bell.not #emoji.brush #emoji.chair #emoji.comet

Sym Raw

 #emoji.bone #emoji.book.open #emoji.briefcase #emoji.bubbles #emoji.coffin #emoji.hook

Sport

Sym Raw

 #emoji.baseball #emoji.chess

Sym Raw

 #emoji.billiards #emoji.climbing

Sym Raw

 #emoji.boxing #emoji.football

Vehicles

Sym Raw

 #emoji.airplane #emoji.boat.sail #emoji.car #emoji.ship

Sym Raw

 #emoji.bike #emoji.boat.speed #emoji.car.front #emoji.motorcycle

Sym Raw

 #emoji.boat #emoji.cablecar #emoji.car.racing #emoji.wheelchair.motor

Building

Sym Raw

 #emoji.castle.eu #emoji.factory #emoji.house

Sym Raw

 #emoji.castle.jp #emoji.hospital #emoji.museum

Sym Raw

 #emoji.circus #emoji.office #emoji.crane

Weather

Sym Raw

Sym Raw

Sym Raw

Sym Raw

	<code>#emoji.cloud</code>
	<code>#emoji.cloud.snow</code>
	<code>#emoji.cloud.sun.hidden</code>
	<code>#emoji.drop</code>

Sym Raw

	<code>#emoji.cloud.dust</code>
	<code>#emoji.cloud.storm</code>
	<code>#emoji.cloud.sun.rain</code>
	<code>#emoji.drops</code>

Sym Raw

	<code>#emoji.cloud.rain</code>
	<code>#emoji.cloud.sun</code>
	<code>#emoji.cloud.thunder</code>
	<code>#emoji.sun</code>

Special

Sym Raw

✓	<code>#emoji.checkmark.heavy</code>
	<code>#emoji.circle.white</code>
✗	<code>#emoji.crossmark</code>
	<code>#emoji.explosion</code>
	<code>#emoji.flag.white</code>
	<code>#emoji.flag.red</code>
	<code>#emoji.globe.eu.af</code>
	<code>#emoji.notes</code>

Sym Raw

	<code>#emoji.checkmark.box</code>
	<code>#emoji.circle.stroked</code>
	<code>#emoji.crossmark.box</code>
	<code>#emoji.fire</code>
	<code>#emoji.flag.black</code>
	<code>#emoji.globe.am</code>
	<code>#emoji.globe.meridian</code>
	<code>#emoji.notes.triple</code>

Sym Raw

	<code>#emoji.circle.black</code>
©	<code>#emoji.copyright</code>
!	<code>#emoji.excl</code>
	<code>#emoji.firecracker</code>
	<code>#emoji.flag.goal</code>
	<code>#emoji.globe.as.au</code>
#	<code>#emoji.hash</code>
*	<code>#emoji.ast</code>

Technology

Sym Raw

	<code>#emoji.battery</code>
	<code>#emoji.brightness.low</code>
	<code>#emoji.calendar</code>
	<code>#emoji.cassette</code>
	<code>#emoji.clamp</code>
	<code>#emoji.clock.alarm</code>
	<code>#emoji.controller</code>
	<code>#emoji.email</code>
	<code>#emoji.folder</code>
	<code>#emoji.hammer</code>
	<code>#emoji.laptop</code>

Sym Raw

	<code>#emoji.battery.low</code>
	<code>#emoji.bubble.speech.l</code>
	<code>#emoji.camera</code>
	<code>#emoji.clip</code>
	<code>#emoji.clock.one</code>
	<code>#emoji.computer</code>
	<code>#emoji.crab</code>
	<code>#emoji.floppy</code>
	<code>#emoji.folder.open</code>
	<code>#emoji.hammer.wrench</code>

Sym Raw

	<code>#emoji.brightness.high</code>
	<code>#emoji.bubble.speech.r</code>
	<code>#emoji.camera.movie</code>
	<code>#emoji.clipboard</code>
	<code>#emoji.clock.two</code>
	<code>#emoji.computermouse</code>
	<code>#emoji.disc.cd</code>
	<code>#emoji.flashlight</code>
	<code>#emoji.gear</code>
	<code>#emoji.headphone</code>

Glossar

FPGA – **Field Programmable Gate Array**: A field-programmable gate array (FPGA) is an integrated circuit designed to be configured by a customer or a designer after manufacturing – hence the term 'field-programmable'. [30](#)

HEI – Haute École d'Ingénierie [30](#)

IT – Infotronics [30](#)

SYND – Systems Engineering [30](#)

Literatur

- [1] S. Zahno u. a., „Dynamic Project Planning with Digital Twin“, *Frontiers in Manufacturing Technology*, Bd. 3, Mai 2023, doi: [10.3389/fmtec.2023.1009633](https://doi.org/10.3389/fmtec.2023.1009633).